

Chapter 4: Stationarity and Invertibility

- The methods applied on previous chapters depends on the underlying assumption that the time-series is stationary.
- However, we can have a presence of time-trend, outliers, structural break and stochastic trend in the data.
- As defined before:
- **Strictly Stationarity**: requires all these moments of the distribution (mean, variance, covariance, kurtosis, etc..) to be independent of time.
- **Weakly stationarity** (or second-order stationary): constant mean, constant variance and the covariance are independent of time (only depends on time difference). A Gaussian process is defined only by its 2 moments, so if it is weakly stationary, then it is strictly stationary.

Stationarity

- **Weakly stationarity:**

- Mean stationary process:

$$E(X_t) = \mu, \text{ for all } t.$$

- Variance stationary:

$$\text{Var}(X_t) = \sigma^2, \text{ for all } t.$$

- Auto-covariance stationary:

$$\text{Cov}(X_t, X_{t+k}) = \text{Cov}(X_{t+a}, X_{t+k+a})$$

does not depend on t , only on the distance between the two observations.

- **Gaussian (normal) processes are defined only by its 2 moments, so if it is weakly stationary, then it is strictly stationary.**

The Importance of Stationarity

- **Stationary processes**: test statistics follow the usual distribution (normal, chi-square).
- **Non-stationary processes**: test statistics do not follow the usual distributions (we will discuss it in chapter 7: unit root tests).
- **Spurious regression problem** (Granger and Newbold-1974): Let's X_t and Y_t be two independent non-stationary variables. If we regress Y_t on X_t , we will find all the statistics significant even when there are not related. The problem here is that because they are not stationary series, the least-square estimates will not be consistent, inferences do not hold. The residuals will not be stationary.

Restrictions on AR parameters to ensure stationarity

- Not all AR processes are stationary. We have seen that in order to be stationary all roots of the lag polynomial, $\Phi(L)$, have to be outside the unit circle. If at least one root is equal to 1, the process is not stationary and it is said to be a unit-root process.
- Example: AR(1):

$$X_t = \beta X_{t-1} + e_t$$

$$(1 - \beta L)X_t = e_t$$

$$\Phi(L) = (1 - \beta L)$$

$$(1 - \beta z) = 0 \Rightarrow z = \frac{1}{\beta} \Rightarrow |\beta| < 1$$

Restrictions on AR parameters to ensure stationarity

- Example: AR(2):

$$X_t = \beta_1 X_{t-1} + \beta_2 X_{t-2} + e_t$$

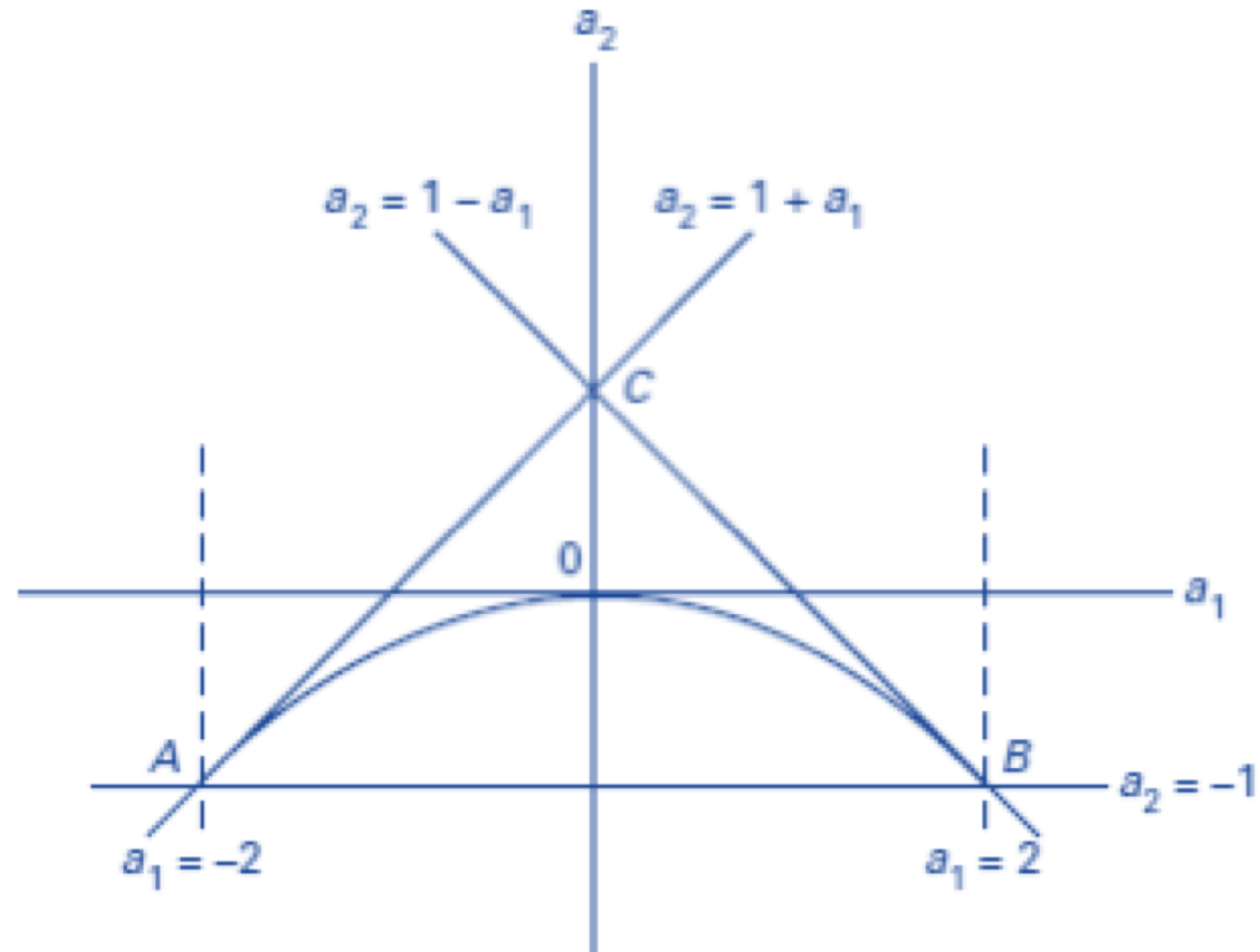
$$(1 - \beta_1 L - \beta_2 L^2)X_t = e_t$$

$$\Phi(L) = (1 - \beta_1 L - \beta_2 L^2)$$

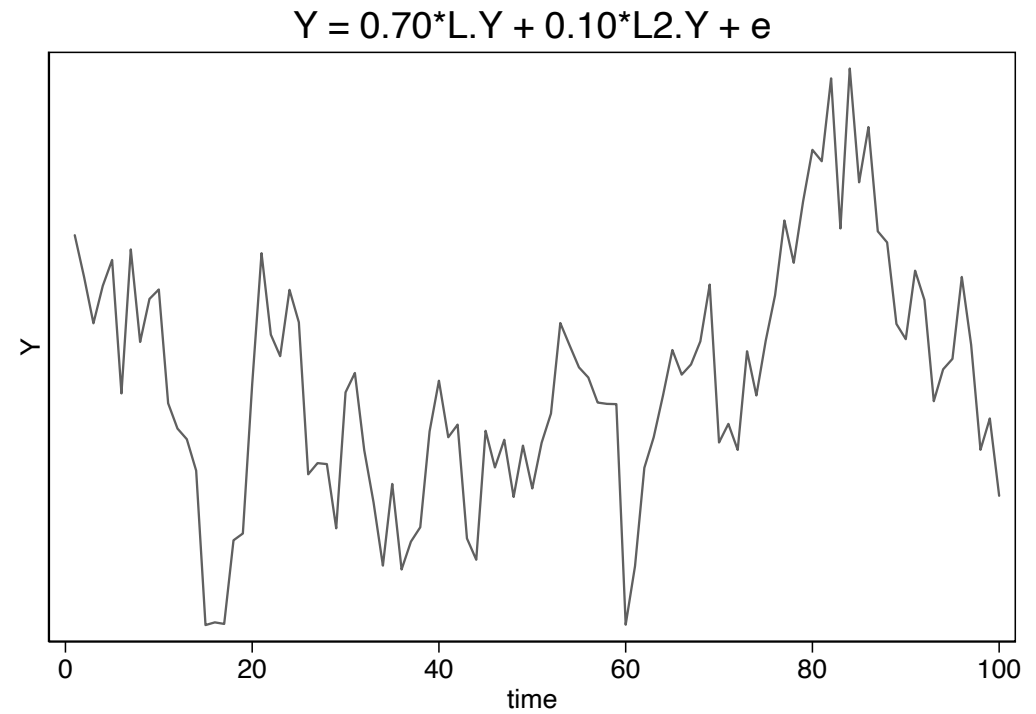
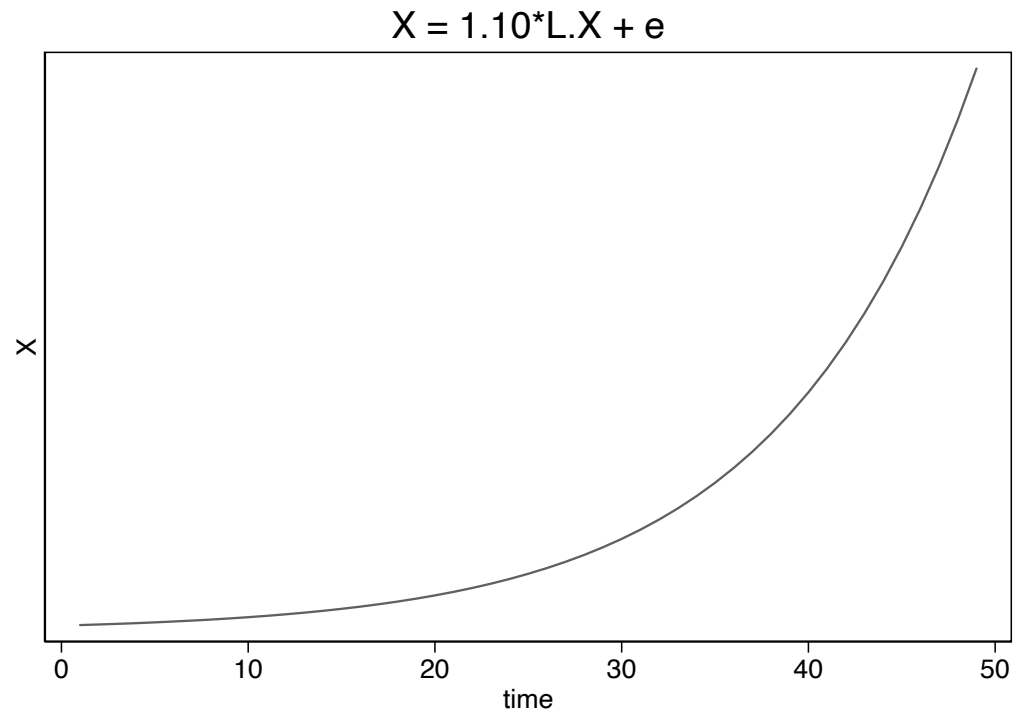
$$(1 - \beta_1 z - \beta_2 z^2) = 0$$

$$\Rightarrow \begin{cases} \beta_1 + \beta_2 < 1 \\ \beta_2 - \beta_1 < 1 \\ |\beta_2| < 1 \end{cases}$$

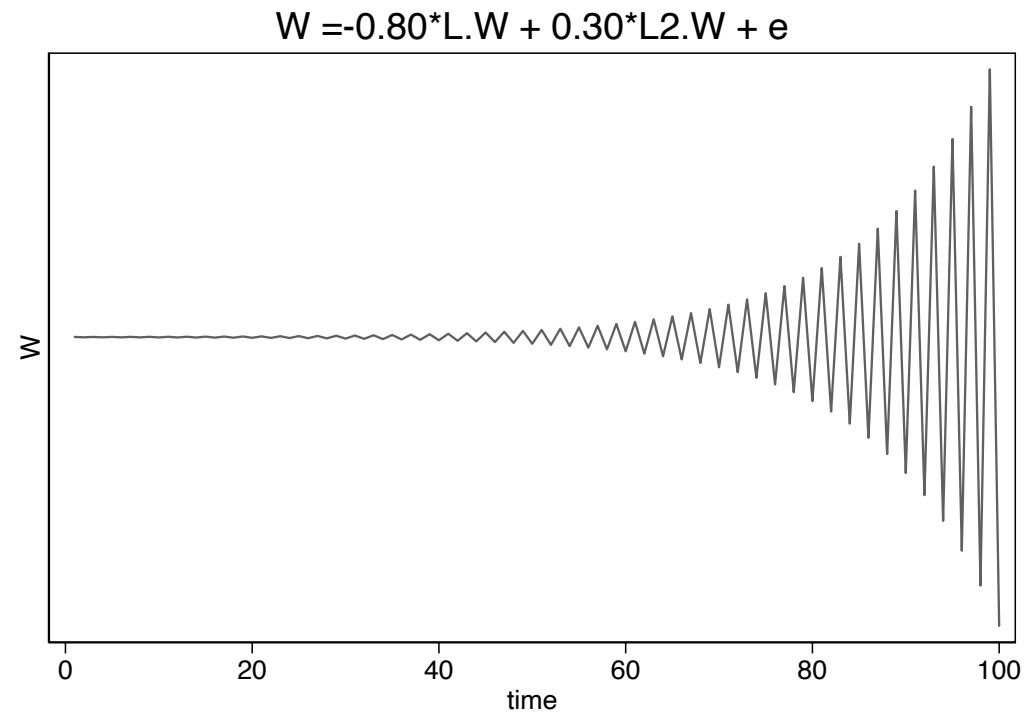
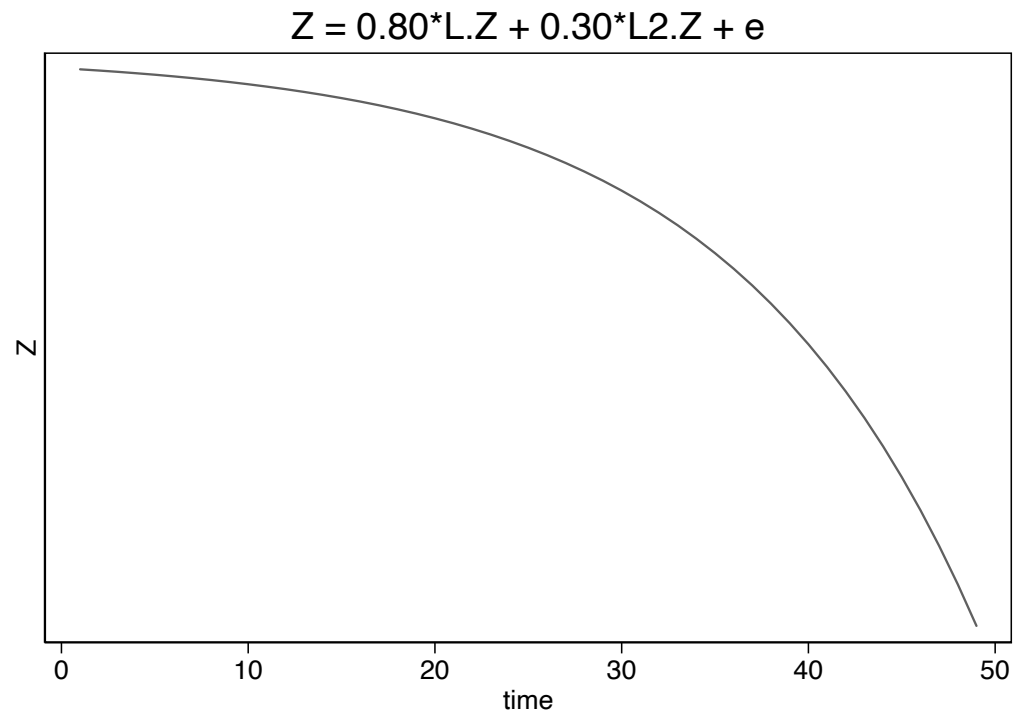
Restrictions on AR parameters to ensure stationarity



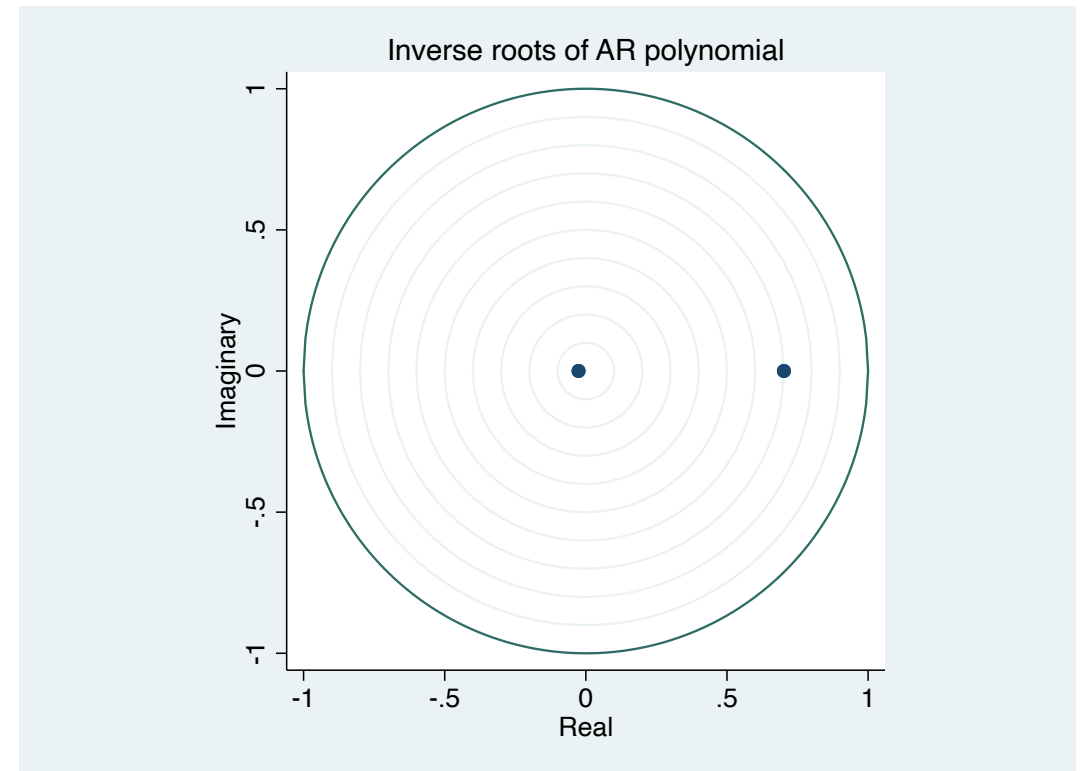
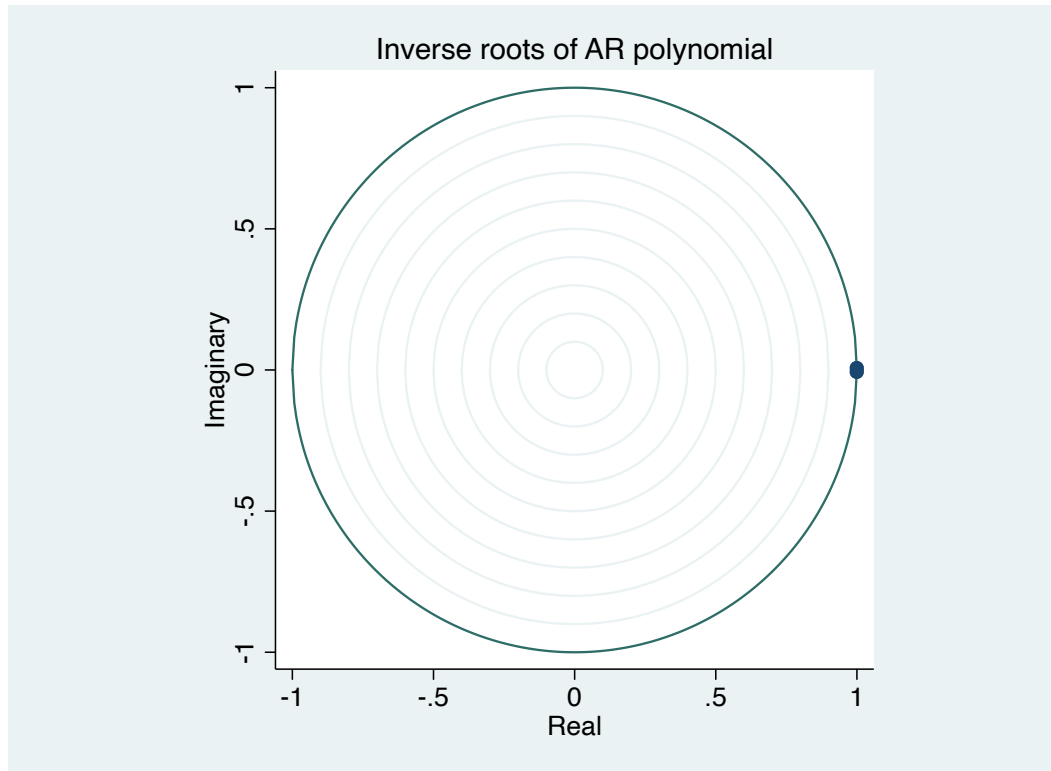
Example: Stationarity conditions



Example: Stationarity conditions



Example of inverse roots of AR(2)
(note: the first AR have “Near unit roots”: necessity of further tests –
Unit Root tests)



Restrictions on MA parameters to ensure invertibility

- Once a AR process is stationary, it is invertible: it can be expressed as an MA(∞)

$$(1 - \beta L)X_t = e_t$$

$$X_t = \frac{e_t}{(1 - \beta L)}, \text{ for } |\beta| < 1$$

$$X_t = (1 + \beta_1 L + \beta_2 L^2 + \dots) e_t$$

$$X_t = 1 + \beta_1 e_{t-1} + \beta_2 e_{t-2} + \dots$$

Restrictions on MA parameters to ensure invertibility

- To ensure invertibility, the roots of the lag polynomial, $\Theta(L)$, has to be outside the unit circle.
- Example MA(1) to AR (∞) :

$$X_t = u_t + \beta u_{t-1}$$

$$X_t = (1 + \beta L)u_t$$

$$\frac{X_t}{\Theta(L)} = u_t, \text{ for } |\beta| < 1$$

$$(1 - \beta_1 L - \beta_2 L^2 - \dots)X_t = u_t$$